A degree-three proof of Campbell's Rogers–Ramanujan evaluation

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Abstract

We prove the conjectured value of $R^5(e^{-\pi\sqrt{16/15}})$. Campbell's proved evaluation at the cubed argument, the classical degree-three modular equation, and an explicit algebraic certificate give the result. A rational bound selects the positive branch.

Task 1: complete proof

For $0 < q < 1$, write

$$R(q) = \cfrac{q^{1/5}}{1 + \cfrac{q}{1 + \cfrac{q^2}{1 + \cdots}}}.$$

The following is Conjecture 1 on page 14 of [1], recorded as MathDB 361714. The published version [2] omits the concluding conjecture without reporting its resolution.

Theorem 1. *Set*

$$a = \frac{8472 + 4875\sqrt{3} - 3770\sqrt{5} - 2175\sqrt{15}}{4}.$$

Then

$$R^5\big(e^{-\pi\sqrt{16/15}}\big) = \sqrt{a^2+1} - a.$$

All radicals denote positive real roots.

We use two previously established inputs, explicitly credited here. The classical degree-three modular equation, displayed as equation (5) in both [1] and [2], is

$$(R(q^3) - R(q)^3)(1 + R(q)R(q^3)^3) = 3R(q)^2R(q^3)^2. \tag{1}$$

Campbell's Theorem 4 in [1], or Theorem 5.2 in [2], proves

$$R^5\big(e^{-\pi\sqrt{48/5}}\big) = \sqrt{b^2+1} - b, \qquad b = \frac{8472 + 4875\sqrt{3} + 3770\sqrt{5} + 2175\sqrt{15}}{4}. \tag{2}$$

Neither input is a new claim of this note.

The algebraic certificate

Define

$$h = \frac{81896 + 47125\sqrt{3} + 36712\sqrt{5} + 21025\sqrt{15}}{602}, \tag{3}$$

$$c = \frac{-8983 - 3250\sqrt{3} + 1651\sqrt{5} + 100\sqrt{15}}{602}, \tag{4}$$

$$t = \sqrt{a^2+1} - a, \qquad s = \sqrt{b^2+1} - b. \tag{5}$$

For indeterminates T, S, let

$$\begin{aligned} Q(T,S) = {} & S^4T - S^3T^4 - 15S^3T^3 - 45S^3T^2 + 15S^3T \\ & - 45S^2T^3 - 528S^2T^2 + 45S^2T \\ & + 15ST^3 + 45ST^2 - 15ST + S - T^3. \end{aligned} \tag{6}$$

Lemma 1. *We have $s = ht + c$ and $Q(t, s) = 0$.*

Proof. Exact expansion in $K = \mathbb{Q}(\sqrt{3}, \sqrt{5})$ gives

$$c = ha - b, \qquad h^2(a^2+1) = b^2 + 1, \qquad Q(T, hT + c) \equiv 0 \pmod{T^2 + 2aT - 1}. \tag{7}$$

These are finite polynomial identities in the displayed constants. To make the third identity reproducible without any root selection or numerical approximation, reduce each factor in the basis $1, T$ using

$$(x_0 + x_1T)(y_0 + y_1T) = (x_0y_0 + x_1y_1) + (x_0y_1 + x_1y_0 - 2ax_1y_1)T.$$

Insert $(0, 1)$ for T and (c, h) for S in the thirteen terms of (6). In the basis $1, \sqrt{3}, \sqrt{5}, \sqrt{15}$ of K, the resulting constant and linear coefficients are respectively

$$(0, 0, 0, 0), \qquad (0, 0, 0, 0).$$

Multiplication in K uses only $(\sqrt{3})^2 = 3$, $(\sqrt{5})^2 = 5$, and $\sqrt{15} = \sqrt{3}\sqrt{5}$. The accompanying standard-library Python certificate implements exactly these operations and also verifies the first two identities.

Every coefficient of h in (3) is positive. Therefore the second identity in (7) gives $\sqrt{b^2+1} = h\sqrt{a^2+1}$. The first identity then gives $s = ht + c$. Since $t^2 + 2at - 1 = 0$, the last identity proves $Q(t, s) = 0$. □

From the norm equation to the correct branch

Put

$$F(V, U) = (U - V^3)(1 + VU^3) - 3V^2U^2 = -V^3 + U - 3V^2U^2 - V^4U^3 + VU^4.$$

The following elementary resultant identity will be used:

$$\prod_{\zeta^5=1} F(V, U\zeta) = Q(V^5, U^5). \tag{8}$$

For completeness, multiplication by $F(V,z)$ in the basis $1, z, z^2, z^3, z^4$ of the algebra $z^5 = S$ has matrix

$$\begin{pmatrix} -V^3 & VS & -V^4S & -3V^2S & S \\ 1 & -V^3 & VS & -V^4S & -3V^2S \\ -3V^2 & 1 & -V^3 & VS & -V^4S \\ -V^4 & -3V^2 & 1 & -V^3 & VS \\ V & -V^4 & -3V^2 & 1 & -V^3 \end{pmatrix}.$$

Its determinant expands to $Q(V^5, S)$; this is also checked by the 120-term determinant expansion in the exact certificate. At $S = U^5 > 0$ the algebra has the five distinct evaluations $z = U\zeta$, so its determinant is the product in (8).

Take $v = t^{1/5} > 0$ and $u = s^{1/5} > 0$. Lemma 1 and (8) show that $F(v, u\zeta) = 0$ for some fifth root of unity ζ. We now prove that $\zeta = 1$.

Rational squaring bounds for the three radicals give $15 < a < 16$ and $8000 < b < 9000$. For example, each radical lies strictly between the following lower endpoint and that endpoint plus 10^{-12}:

$$\frac{1732050807568}{10^{12}}, \quad \frac{2236067977499}{10^{12}}, \quad \frac{3872983346207}{10^{12}} \quad \text{for } \sqrt{3}, \sqrt{5}, \sqrt{15}, \text{ respectively.}$$

Hence

$$0 < t = \frac{1}{\sqrt{a^2+1}+a} < \frac{1}{30} < \left(\frac{3}{5}\right)^5, \qquad 0 < s < \frac{1}{16000} < \left(\frac{1}{5}\right)^5.$$

Thus $0 < v < 3/5$ and $0 < u < 1/5$. For a nontrivial fifth root of unity, $|\operatorname{Im}\zeta| > 1/2$. The imaginary part of $F(v, u\zeta) = 0$, followed by the triangle inequality and division by u, would imply

$$\frac{1}{2} < |\operatorname{Im}\zeta| \le vu^3 + v^4u^2 + 3v^2u < \frac{3531}{15625} < \frac{1}{2},$$

which is impossible. Consequently $F(v, u) = 0$.

Identification with the continued fraction

Set $q = e^{-\pi\sqrt{16/15}}$. Then $q^3 = e^{-\pi\sqrt{48/5}}$, so (2) identifies $u = R(q^3)$. For this fixed $u > 0$, the polynomial

$$F(V, u) = -u^3V^4 - V^3 - 3u^2V^2 + u^4V + u$$

has exactly one sign change in its nonzero coefficient sequence. Descartes' rule therefore gives at most one positive root. We have already proved that $v > 0$ is such a root. The positive number $R(q)$ is also a root by (1). Thus $R(q) = v$ and $R(q)^5 = t$, proving the theorem. □

Tasks 2–5: proposed follow-up directions

These are further questions, not additional solved problems.

Task 2: Determine the irreducible minimal polynomial of $R(e^{-\pi\sqrt{16/15}})$ and the Galois group of its splitting field, including the role of the field $K(\sqrt{a^2+1})$ used above.

Task 3: Classify the pairs $a, b \in \mathbb{Q}(\sqrt{3}, \sqrt{5})$ for which $\sqrt{b^2+1} - b = h(\sqrt{a^2+1} - a) + c$ with h, c in the same field and the corresponding degree-three modular equation holds.

Task 4: Determine the full region of positive (v, u) for which $Q(v^5, u^5) = 0$ forces $F(v, u) = 0$, sharpening the sufficient condition $vu^3 + v^4u^2 + 3v^2u < \sin(\pi/5)$.

Task 5: Use the degree-two and degree-five modular equations to obtain explicit radical expressions at the arguments q^2 and q^5, starting from the value proved here, with exact branch certificates.

Verification and attribution. The proof uses the two credited published inputs (1) and (2). Its new algebra and branch selection are exact. The accompanying verifier checks four groups: radical correlation, polynomial reduction, the norm determinant, and rational branch bounds. The authoring agent reviewed this proof; independent peer review and proof-assistant certification are not claimed.

References

[1] J. M. Campbell, *Applications of the icosahedral equation for the Rogers–Ramanujan continued fraction*, arXiv:2402.02546v1 (2024), `https://arxiv.org/abs/2402.02546`.

[2] J. M. Campbell, *Applications of the icosahedral equation for the Rogers–Ramanujan continued fraction*, J. Korean Math. Soc. **63** (2026), 1013–1025, `https://doi.org/10.4134/JKMS.j250500`.